

CV Practical Assignment

1. Write a Program for any image. Crop a particular portion of image. Change the intensity value of that particular section.
2. Enter any point in the image and mark all the 4 neighbours, 8 neighbours.
3. Write a program to calculate all the three distance measures. Show that euclidean distance is minimum of all the three.
4. Perform[image negation, log transformation, gamma transformation, piecewise-linear transformation] using both built-in and custom function.
5. Perform Bit Plane Slicing
6. Program to perform image smoothening using filters of different sizes with hard coding as well as opencv
7. Write hard coded program for 2D convolution for image averaging (simple and weighted) and median.
8. Perform image smoothening (noise reduction) using:
 - a) Mean Filter(Average Filter)
 - b) Gaussian Blur
 - c) Median Filter
 - d) Weighted Mean Filter
 - e) Min and Max Filters
9. Write a program to detect edges in an image using OpenCV:
 - a) Filter2D with Laplacian Kernel
 - b) Laplacian Method in OpenCV
 - c) Sobel Operator